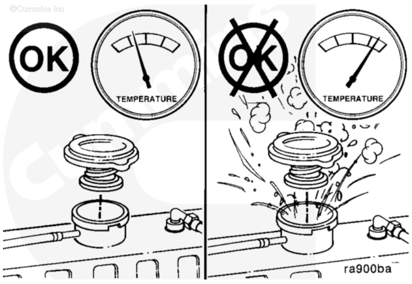
Trainings ▾

Drain

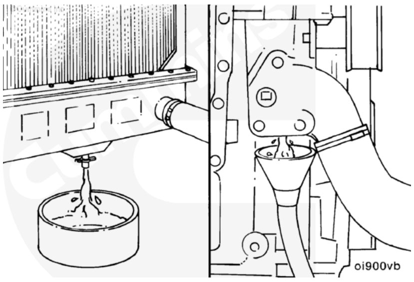
⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.



⚠ WARNING ⚠

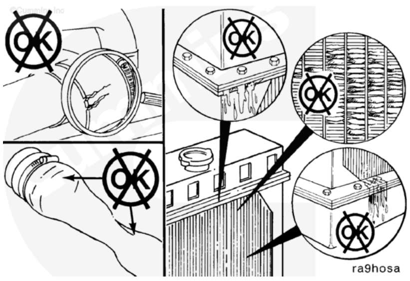
Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.



Drain the cooling system by opening the drain valve on the radiator and removing the plug in the bottom of the water inlet. A drain pan with a capacity of 19 liters [5 gal] will be adequate in most applications.

Check for damaged hoses and loose or damaged hose clamps. Replace as required.

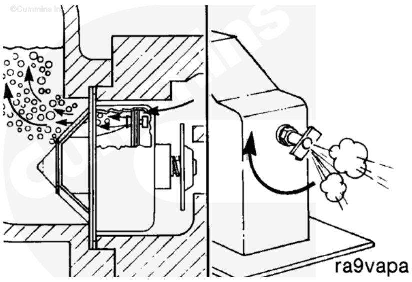
Check the radiator for leaks, damage, and buildup of dirt. Clean and replace as required.



Flush

⚠ CAUTION ⚠

During filling, air must be vented from the engine coolant passages. The air vents through the jiggle pin openings to the top radiator hose and out the fill opening. Additional venting is provided for engines equipped with an aftercooler. Open the petcock during filling.



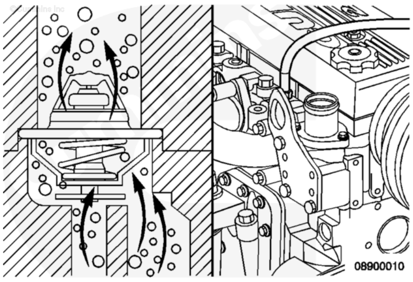
For front gear train engines with aftercoolers, open the petcock during filling for additional venting.

For front gear train engines without aftercoolers, the air vents through the jiggle pin openings in the thermostat to the top radiator hose and out the fill opening.

This provides adequate venting for a fill rate of 19 liters per minute [5 U.S. gallons per minute].

⚠ CAUTION ⚠

The system must be filled properly to prevent air locks. During filling, air must be vented from the engine coolant passages. Wait 2 to 3 minutes to allow air to be vented then add mixture to bring the level to the top.



For rear gear train engines, a dearation port next to the water outlet connection vents air to the top tank of the cooling system.

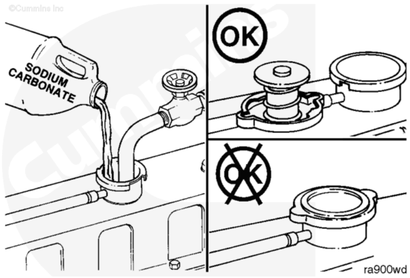
This provides adequate venting for a fill rate of 19 liters per minute [5 U.S. gallons per minute].

Note : An alternative to using sodium carbonate, as outlined in this procedure, is to use RESTORE™.

RESTORE™ is a heavy-duty cooling system cleaner that removes corrosion products, silica gel, and other deposits. The performance of RESTORE™ is dependant on time, temperature, and concentration levels. An extremely scaled or flow-restricted system, for example, can require higher concentrations of cleaners, higher temperatures, or longer cleaning times or the use RESTORE Plus™. Up to twice the recommended concentration levels of RESTORE™ can be used safely. RESTORE Plus™ **must** be used only at its recommended concentration level. Extremely scaled or fouled systems can require more than one cleaning.

⚠ CAUTION ⚠

Do not install the radiator cap. The engine is to be operated without the cap for this process.

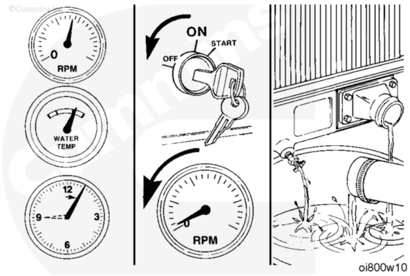


Fill the system with a mixture of sodium carbonate and water (or a commercially available equivalent).

Note : Adequate venting is provided for a fill rate of 19 liters per minute [5 U.S. gallons per minute].

⚠ WARNING ⚠

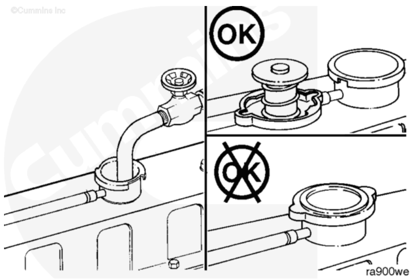
Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.



Operate the engine for 5 minutes with the coolant temperature above 80°C [176°F].

Shut the engine OFF, allow to cool to 50°C [120°F], and drain the cooling system.

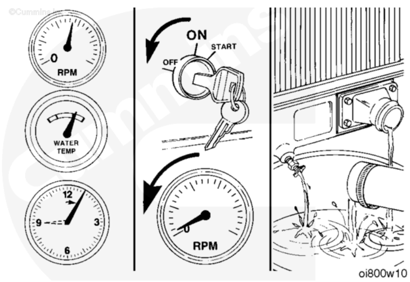
Fill the cooling system with clean water.



Note : Be sure to vent the engine and aftercooler, if equipped, for complete filling.

Note : Do not install the radiator cap.

Operate the engine for 5 minutes with the coolant temperature above 80°C [176°F].



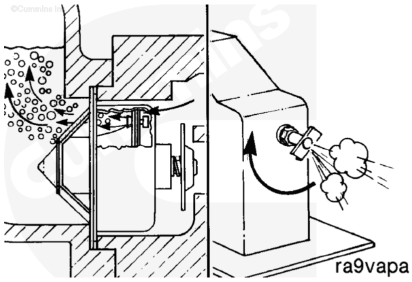
Shut the engine OFF, allow to cool to 50°C [120°F], and drain the cooling system.

Note : If the water being drained is still dirty, the system must be flushed again until the water is clean.

Fill

⚠ CAUTION ⚠

During filling, air must be vented from the engine coolant passages. The air vents through the jiggle pin openings to the top radiator hose and out the fill opening. Additional venting is provided for engines equipped with an aftercooler. Open the petcock during filling.



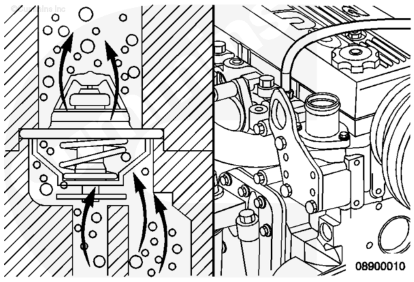
For front gear train engines with aftercoolers, open the petcock during filling for additional venting.

For front gear train engines without aftercoolers, the air vents through the jiggle pin openings in the thermostat to the top radiator hose and out the fill opening.

This provides adequate venting for a fill rate of 19 liters per minute [5 U.S. gallons per minute].

⚠ CAUTION ⚠

The system must be filled properly to prevent air locks. During filling, air must be vented from the engine coolant passages. Be sure to open the petcock on the aftercooler for aftercooled engines. Wait 2 to 3 minutes to allow air to be vented; then add mixture to bring the level to the top.

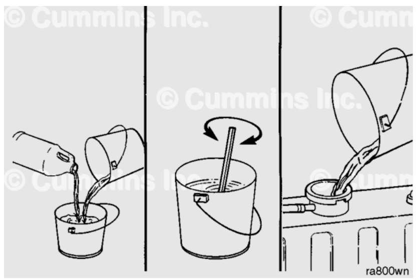


For rear gear train engines, a dearation port next to the water outlet connection vents air to the top tank of the cooling system.

This provides adequate venting for a fill rate of 19 liters per minute [5 U.S. gallons per minute].

⚠ CAUTION ⚠

Never use water alone for coolant. Damage from corrosion can be the result of using water alone for coolant.

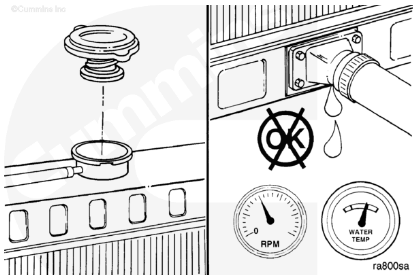


Use a mixture of 50 percent water and 50 percent ethylene glycol or propylene glycol antifreeze to fill the cooling system. See Cummins Coolant Requirements and Maintenance, Bulletin 3666132 for engine coolant specifications.

Refer to Procedure Procedure 018-018 in Section V (/qs3/pubsys2/xml/en/procedures/40/40-018-018-tr.html) for system capacity.

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [122°F]. Failure to do so can cause personal injury from heated coolant.



Install the pressure cap. Operate the engine until it reaches a temperature of 80°C [180°F], and check for coolant leaks.

Check the coolant level again to make sure system is full of coolant, or that the coolant level has risen to the hot level in the recovery bottle on the system, if equipped.

Last Modified: 08-Apr-2015